

“This Isn’t Chatbot AI. It Is AI For SENSOR DATA From Space.”

AI is making satellites smarter. But how does this benefit humanity on Earth, and what problems can it solve? In a conversation with Nidhi Agarwal of EFY, Vishesh Vatsal from SkyServe explains how AI operates in orbit—and why it matters.

VISHESH VATSAL
CO-FOUNDER AND CTO, SKYSERVE



What are you developing to make the satellites smarter?

We build edge computing software infrastructure. This is a system that lets you define what a satellite should do. Satellites have payload computers. Our software runs on these computers, allowing applications to be set up from Earth and executed on the satellite. This setup lets the satellite do computing tasks in space. You can build models on Earth, send them to the satellite, run them there, and get the results back. It is modular and can scale to support a large number of satellites in a network.

Can you briefly say what AI and deep learning do in your innovation?

While many associate AI with chatbots and generative tools, our focus is AI for processing satellite sensor data. This includes analysing imagery—such as identifying runway damage during Operation Sindoor—using computer vision and remote sensing models.

But it is not just images. Satellites also produce other types of data, like time-series data and health telemetry. We use AI to find patterns or changes in that data. The models we use include computer vision, deep learning, time-series analysis, anomaly detection, and, occasionally, LLM infrastructure, although that is not always necessary.

What design problems did you face?

One major challenge with using AI/ML models on edge computers in satellites is the limited power and computing capacity. Unlike cars or desktop computers, satellites cannot afford to use high power or memory. We must therefore

be precise about memory requirements and power consumption. To meet these constraints, we use model compression and other optimisation techniques to ensure the models perform effectively within tight limits.

How is power managed during AI work in space?

Power consumption management on satellites is handled through careful coordination with the mission operations team. Based on this, they define power usage profiles and operation schedules to ensure that everything stays within the satellite’s available power budget. This helps define the performance limits within which we must operate. That said, with the rapid advancement in low-power computing hardware, this challenge may soon fade. As electronics become more power-efficient, we expect power constraints to become less of an issue in future satellite missions.

How do you update the AI model after launch? Is there a way to retrain or tune it remotely?

We use an over-the-air update system. Instead of sending the entire model again, we just send the difference between the current model on the satellite and the new one.

What makes it hard to run a geo-AI model in space vs on Earth?

One major issue is that most models are trained on clean, high-quality datasets from Earth, but imagery captured in space can be quite different. Another big challenge is creating

analysis-ready data. That involves tasks such as geo-referencing and adding contextual information with high accuracy, which is very challenging to do onboard. These are complex problems, but we are working on them and expect they will be gradually solved over time.

Who are your target customers?

Today, earth observation has two main types of users—half are government users like disaster response teams, and the other half are commercial users. Governments primarily use it to gain faster insights for situations such as emergencies. Commercial users include those in shipping, insurance, or disaster management.

How do you see edge AI growing in future Earth missions with many sensors and data types?

With the increasing number of sensors on satellites, we are seeing a rapid rise in both the quality and quantity of data being generated in space. Quality refers to higher spectral and spatial resolution, as well as more sensitive sensors, whereas quantity refers to the sheer volume of data produced. To handle this, edge AI will need to evolve, adapting to process more complex data directly on board. In future, we will be running advanced models in space that were previously only feasible in large industrial setups on Earth. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/interviewOct25>

